

BOEC ENGINEERING CONSULTANTS  
SANAS Accredited Calibration Laboratory  
No. CAL-2024-07  
ISO/IEC 17025:2017 & ISO4037-3  
Compliant

CERTIFICATE NO: BOEC-CAL-2024-084  
Date: 2026-09-12  
Template: MCC15-07 v2.1  
DIGITALLY SIGNED

1. Client Information

Client Name: Anglo American Radiation Services  
Address: Koeberg Rd, Melkbosstrand  
Contact: Radiation Protection Manager

2. Instrument Under Test

Model: Ludlum Model 14C Survey Meter  
Serial Number: SN-2026-9941  
Procedure: CP-02-08 / CP-03-07

3. Environmental Conditions (ISO4037-3 Clause 6.2)

| Parameter         | Measured Value | Required Limits     | Status |
|-------------------|----------------|---------------------|--------|
| Temperature       | 21.8 °C        | 18.0 °C – 24.0 °C   | PASS   |
| Relative Humidity | 42.5 %RH       | 30.0 %RH – 60.0 %RH | PASS   |
| Pressure          | 85.2 hPa       | 860 hPa – 1060 hPa  | PASS   |

4. Calibration Irradiation Results (Cs-137 Gamma Source)

| Nominal Dose Rate | Measured Dose Rate | Response Factor | Uncertainty (k=2) | Compliance |
|-------------------|--------------------|-----------------|-------------------|------------|
| 0.500 mSv/h       | 0.512 mSv/h        | 1.024           | ± 4.5%            | PASS       |
| 2.000 mSv/h       | 2.043 mSv/h        | 1.022           | ± 4.3%            | PASS       |
| 10.00 mSv/h       | 10.18 mSv/h        | 1.018           | ± 4.2%            | PASS       |

5. Metrological Traceability & RSA Digital Signature Seal

Metrological Traceability: Traceable to SI units held at NMISA  
(Cert: 2023-11-0445).  
Technical Signatory: T. Dlamini (SANAS Signatory)  
Approval Date: 2026-09-12

DIGITAL SIGNATURE VERIFIED  
Signature: BOEC-RSA2048-SIG-EBE32FA76FD49CF3  
SHA256: ebe32fa76fd49cf3ab72...  
Status: RELEASED / PASSED